SMART INDIA HACKATHON 2026

AI-POWERED CRIMINAL NETWORK ANALYSIS SYSTEM

A Software-Based Platform Proposal for Ministry of Home Affairs (MHA)

Category:	Software Edition (100% Code-based)
Problem Statement ID:	SIH26189
Focus Domain:	AI/ML, Graph Analytics, Cyber Investigations
Primary Tech Stack:	FastAPI, React, Neo4j, Vis.js, NLP (spaCy)

Prepared by: NIT Delhi Hackathon Team
AI Mentor: Antigravity Assistant
Date: August 30, 2026

1. Executive Summary & Working Mechanics

Modern organized crime syndicates operate across jurisdictions, masking their actions through a complex web of throwaway phones, decoy bank accounts, and localized relationships. While law enforcement agencies (LEAs) collect copious amounts of raw data—such as First Information Reports (FIRs), Call Detail Records (CDRs), and Bank Statements—these records are stored in siloed, unstructured formats. Investigators are forced to manually correlate these scattered files on paper charts or disjointed Excel files.

The **AI-Powered Criminal Network Analysis System** automatically reads these heterogeneous files, extracts names, aliases, locations, phone numbers, and financial details, and fuses them into a central **Graph Database**. It then runs mathematical graph algorithms to reveal key organizational patterns.

How the Ingestion-to-Visualization Pipeline Works:

- **1. File Upload:** The investigator uploads raw, multi-format files (PDF case reports, CSV phone logs, Excel bank sheets).
- **2. Multi-lingual AI Text Parser (NLP):** An NLP pipeline identifies entities. If reports are in regional Indian languages (e.g., Hindi), a translation bridge translates/transliterates them to standardized English.
- **3. Entity Resolution:** The system resolves duplicate entities (e.g., if a suspect 'Rajesh' uses the same phone number as 'Rajesh Kumar' in a different report, they are merged into one node).
- **4. Graph Relational Database (Neo4j):** Nodes and relationships (e.g., 'A called B', 'B transferred funds to C') are stored in a graph network.
- **5. Graph Analytics:** Algorithms compute PageRank/Centrality scores to find 'hubs' (ring leaders) and trace Dijkstra's Shortest Path to link remote suspects.
- **6. Interactive Dashboard (Vis.js):** The network is rendered as an interactive, draggable visual interface with search, filters, and timeline playbacks.

2. How This System Compares to Existing Methods

To understand the value of this system, we must compare it with the current 'status quo' in local Indian police stations as well as high-end enterprise software like Palantir and IBM i2 Analyst's Notebook.

Criteria	Current Police Method	Enterprise Tools (Palantir/i2)	Our Proposed System
Input Processing	Manual entry / physical highlighters on paper reports.	Ingestion of clean, pre-structured databases.	**Automated AI parsing** of raw, unstructured text files.
Data Fusion	CDRs, bank accounts, and case texts remain in siloed sheets.	Requires custom enterprise database connectors.	**Unified Ingestion**: Integrates text, phone logs, and finance automatically.
Language Support	Relies entirely on bilingual investigators.	Strictly English/major global languages.	**Indian Vernacular**: NLP transliteration and translation.
Cost & Setup	Free, but highly time-consuming and error-prone.	Extremely expensive licenses, weeks of training.	**Open-source**, browser-based, zero installation load.

Key Advantages for SIH Evaluation:

1. **Zero-Setup Compatibility:** During the 36-hour hackathon and live presentation, judges will not wait for a full database setup. Our backend will implement a **dual-mode database structure**. It runs in Neo4j if available, but automatically falls back to a lightweight in-memory Graph structure if Neo4j is offline. This guarantees a crash-free demo.
2. **Dynamic Temporal Timeline:** Criminal rings operate dynamically. Adding a date-slider on the frontend allows judges to see how the connections expanded over time, transforming a static chart into an active investigative playback tool.

3. 36-Hour Hackathon Development Roadmap

To successfully deliver this system as a high-functioning prototype during the grand finale, we will divide the work into modular steps:

Hours	Target Module	Key Deliverables
Hours 0 - 6	Environment & Scaffolding	Set up FastAPI, create React/Tailwind frontend shell, integrate Vis.js visual library.
Hours 6 - 12	AI & NLP Parser	Write the spaCy NER extractor to parse raw text and pull names, phone numbers, and locations.
Hours 12 - 18	Graph Engine Integration	Set up the NetworkX / Neo4j services. Connect extracted entities into nodes and directed edges.
Hours 18 - 24	Graph Calculations API	Implement PageRank (gang leaders), Dijkstra Shortest Path (multi-hop links), and Dijkstra API triggers.
Hours 24 - 30	Interactive Dashboard	Bind frontend UI to backend APIs. Render visual graph, add search highlights, timeline slider.
Hours 30 - 36	Polishing & Mock Data	Ingest realistic case data (FIRs/CDRs/Transactions), test edge cases, configure auto-run script.

By utilizing this roadmap, the team ensures a highly polished, working prototype that matches all evaluation criteria set by the Ministry of Home Affairs for the SIH 2026 hackathon.